

## WindowWidth [SimTalk]

Sets the width in pixels with which *Plant Simulation* opens the display window of the *HtmlReport* designated by <Path>.

### Syntax

```
<Path>.WindowWidth:integer
```

### Assignment Value

You can assign a value of data type *integer*.

### Example

```
MyHtmlReport.WindowWidth := 800 // pixels
```

### SimTalk

[WindowWidth \[SimTalk\]](#)

[WindowHeight \[SimTalk\]](#)

See also

[Show Report](#)

## Dialog

### Dialog [object]

Use the object *Dialog* for designing a dialog box for an object which is similar to the built-in dialog boxes.

### Image



### Description

With the object *Dialog* you can:

- Provide the user with a simple user interface for complex simulation models other users work with. This way you can ask the user to type in information which *Plant Simulation* needs to carry out a task.

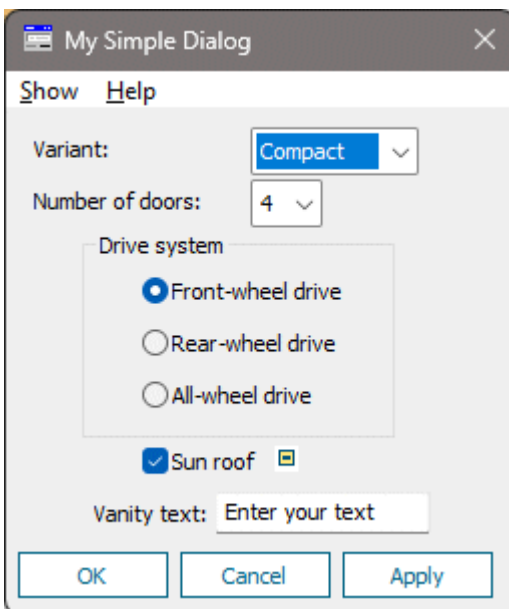
- Prevent the user from manipulating a *Frame* in which you modeled a complex machine, etc. To do so, insert a *Method* object as an **Open Control** into the *Frame*. Double-clicking the *Frame* will then not open it any longer, but call the method. The method in turn executes the **parameter** that you typed in, i.e., it opens the dialog, where the user can then select the settings.

Each *Dialog* object manages a single dialog box. If you need more than one user-defined dialog boxes in your simulation model, insert as many *Dialog* objects as you need.

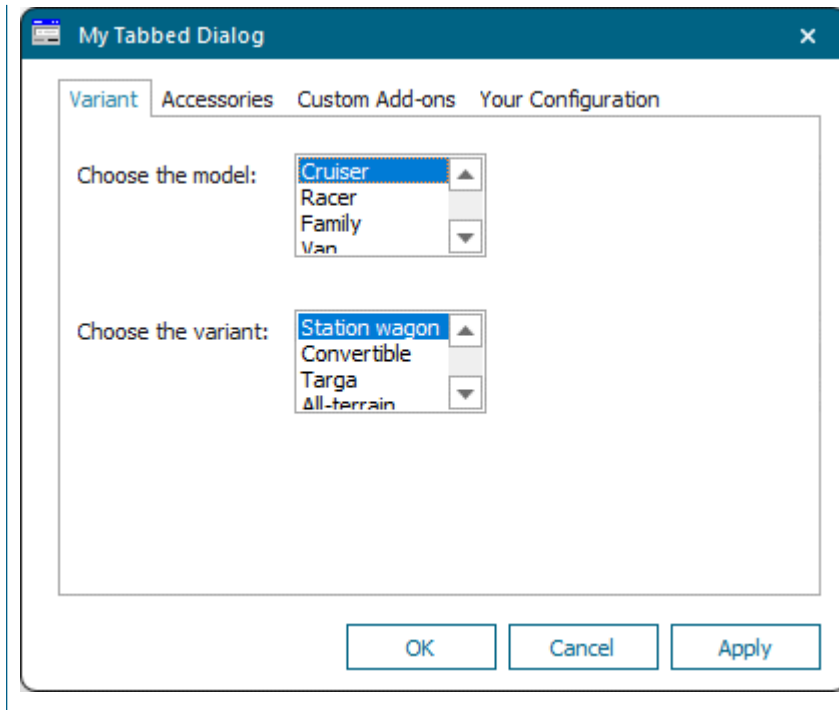
- To create and insert a dialog item, click the right mouse button on the tab **Elements** and select a dialog item on the context menu, compare [Insert a Dialog Item into Your Dialog](#).
- To show the dialog you are creating, click [Show Dialog](#).

### Note

The **display window** of the object *Dialog* that contains menus does not use the standard Siemens PLM theme, but your Windows theme.



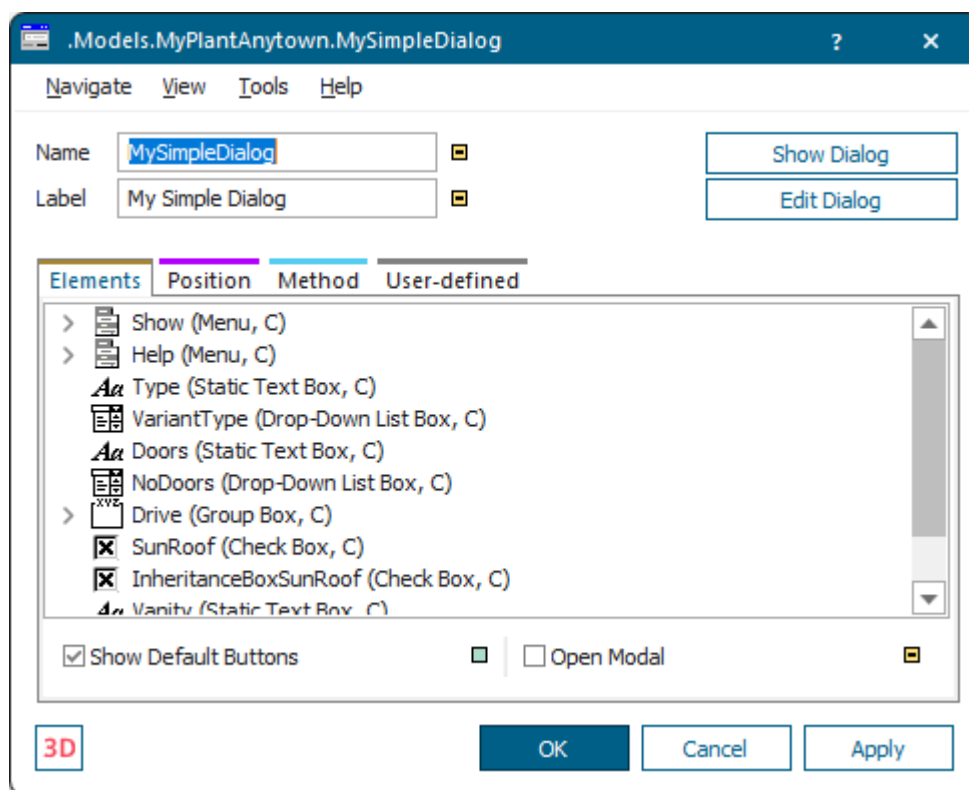
The **display window** of the object *Dialog* that does not contain menus uses the standard Siemens PLM theme.



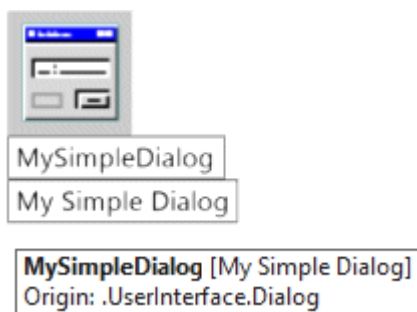
- To edit the position of a dialog item in the dialog you are creating, click **Edit Dialog**, select it and drag the mouse to a new location.

**Note**

The **dialog box** of the object *Dialog* uses the standard Siemens PLM theme.



To show a **tooltip** with information about the *Dialog*, hover with the mouse over it.



To change the length of the graphic and the anchor points of the *Dialog*, click **Show Manipulators** on the **Edit** ribbon tab or press **M** on the keyboard.



### Add the Object to the Simulation Model

To add the object *Dialog* to your simulation model, click **Manage Class Library** > **Basic Objects** > **UserInterface** > **Dialog** on the **Home** ribbon tab.



Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

We created the dialog boxed of all the **Tools** with the *Dialog*.

See also

[Set Parameters for Objects in Your Own Dialog](#)

## Dialog Box of the Dialog

### Dialog Box of the Object Dialog

Double-click the icon of the *Dialog* to open its dialog box.

#### Edit Simulation Properties

In the dialog box you can change the **simulation properties** of the object. The shared properties are described under [Dialog Items of the Objects](#).

#### Edit Animation Properties

To edit the **3D properties** of the object in the dialog box [Edit 3D Properties](#):

- Click the button  **Edit 3D Properties** in the lower left corner of the simulation properties dialog box.
- Select the object in the model and press the **spacebar**.

To manipulate the graphic of the object, click [Show Manipulators](#) on the **Edit** ribbon tab or press **M** on the keyboard.

See also

[Show Dialog \[Dialog\]](#)

[Edit Dialog \[described\]](#)

[Label \[text box\] - Dialog](#)

## Label [text box] - Dialog

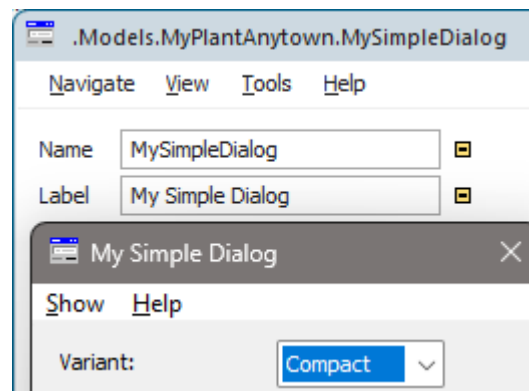
Type in a **Label** for the dialog you create. As opposed to the name, you can type in any expression of your choice as the label, including special characters and blank spaces.

Label:  

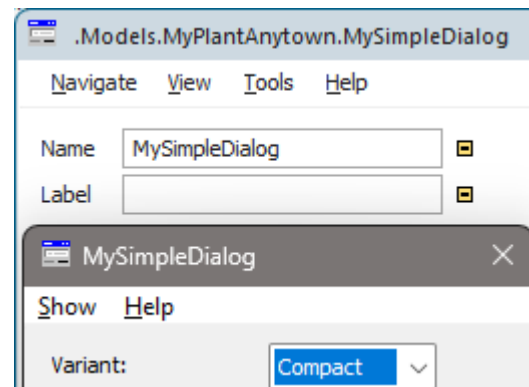
### Remarks

You can:

- Type in a label. *Plant Simulation* then shows that **Label** in the title bar of the dialog you created.



- Leave the text box **Label** empty. *Plant Simulation* then shows the **Name** of the *Dialog* object in the title bar of the dialog.

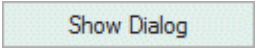


## SimTalk

### Label [SimTalk]

## Show Dialog [Dialog]

To show the dialog with the dialog items you created, click this button.



Show Dialog

### Remarks

The *Dialog* executes the **Callback Method** with the parameters you typed into the text boxes for **Argument for Open**, **Argument for Apply** or **Argument for Close**. This way you can detect if they do what you programmed them to do.

#### Note

*Plant Simulation* shows the dialog with the settings you select under **Personalization > Font** as well as with the settings you select under **Start > Settings > System > Display > Scale & layout**. If you change these settings, the dialog scales text, while the images always retain their size in pixels. For this reason it can happen that text overlaps the image. We recommend to test your dialogs with different display settings to make sure that everything works as you expect it.

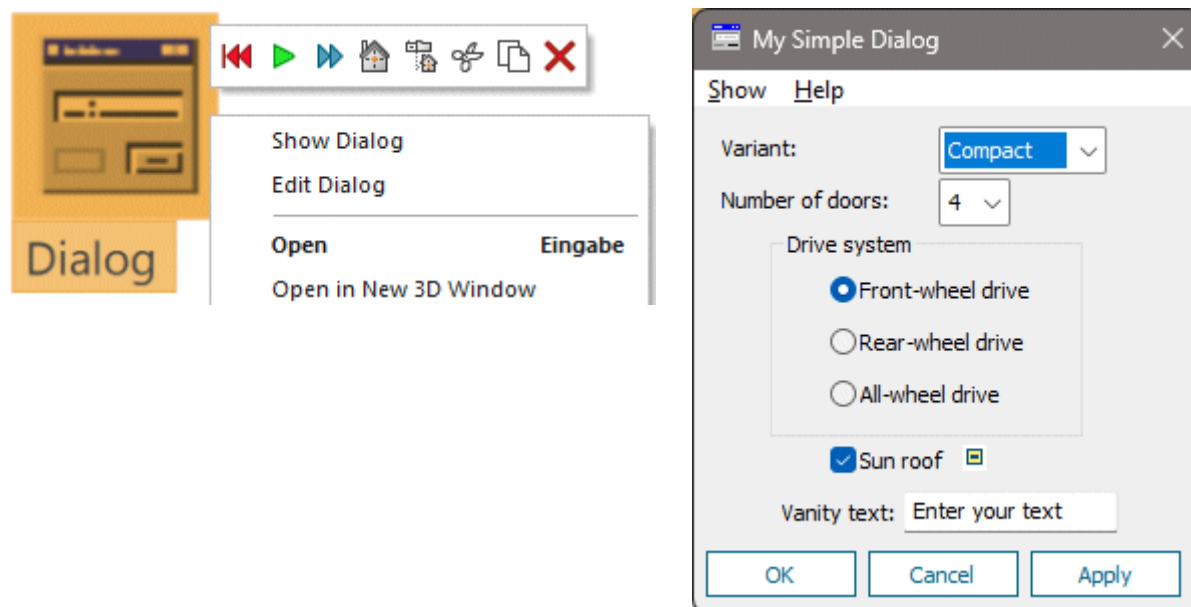
#### Note

Dialog items which you create may be hidden by other dialog items if they have the same **X** and/or **Y** coordinates.

#### Note

As opposed to the object *Dialog*, which previous versions of *Plant Simulation* provided, which used one **Callback Method** per action, the current version of *Plant Simulation* uses a **single callback method** for defining **all** of the actions.

Instead, you can also right-click the *Dialog* object and select **Show Dialog** on the context menu in the *Frame*.



### Note

The **display window** of the object *Dialog* that contains menus does not use the standard Siemens PLM theme, but your Windows theme.

## SimTalk

[openDialog \[SimTalk\] - Dialog](#)

[close \[SimTalk\] - Dialog](#)

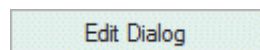
[closeDialog \[SimTalk\] - Dialog](#)

[open \[SimTalk\] - Dialog](#)



## Edit Dialog [described]

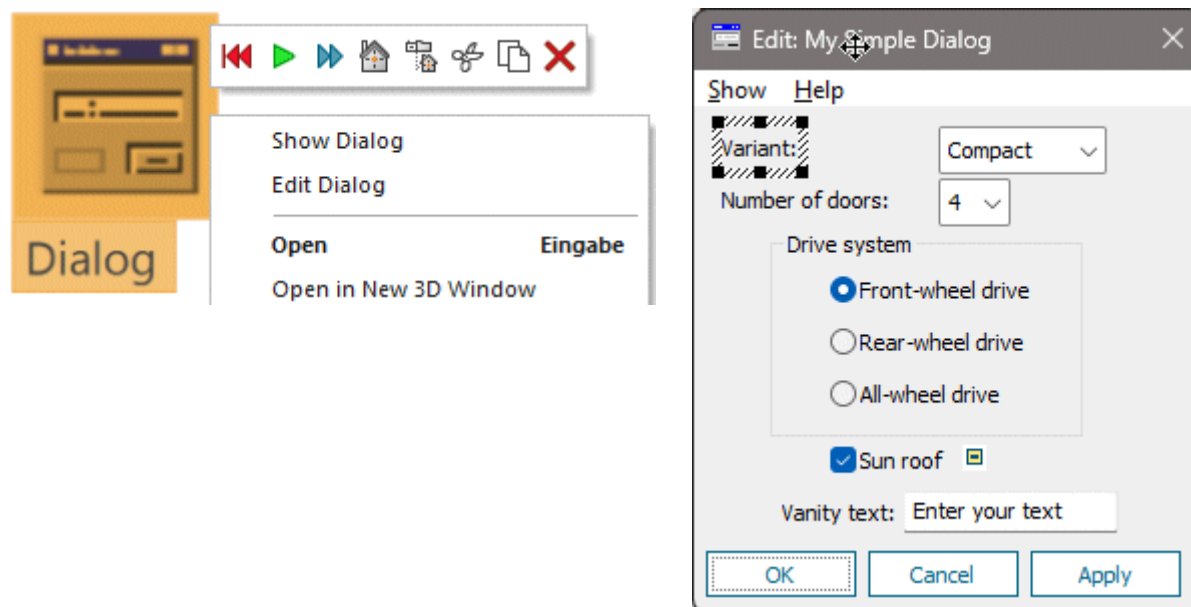
To edit a dialog item which you created in the *Dialog*, click this button. The *Dialog* then selects the dialog item in the display window, and you can move it by dragging it to another position.



### Remarks

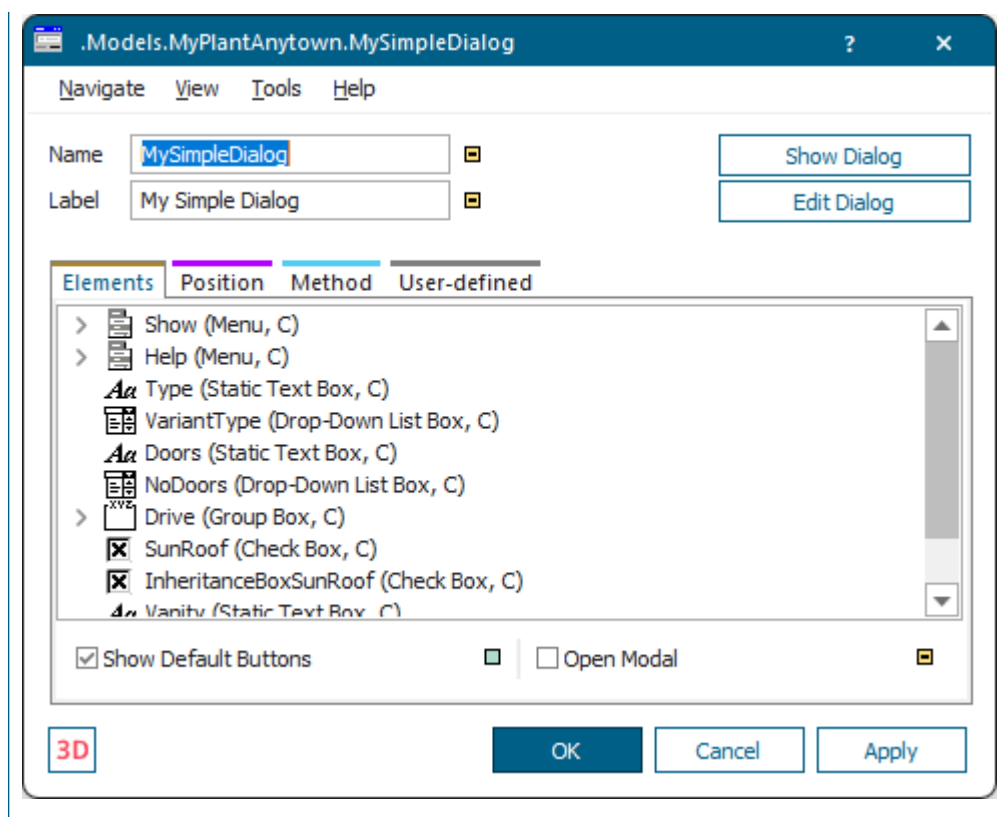
To edit its properties, click it with the right mouse button and select **Open** on the context menu.

Instead, you can also right-click the *Dialog* object and select **Edit Dialog** on the context menu in the *Frame*.



### Note

The **dialog box** of the object *Dialog* uses the standard Siemens PLM theme.



## Tab Elements

## Tab Elements

The tab **Elements** provides a context menu with the commands for inserting dialog items into the dialog you are creating.

### Remarks

To [Insert a Dialog Item into Your Dialog](#), click into the text box with the right mouse button and select the respective item on the context menu.

You can also select to show or hide the **Default Buttons** (**OK**, **Cancel** and **Apply**) and to open the dialog you create **Modal**, meaning that the user cannot open another dialog until he closes the present dialog. Then the user can only open another dialog after he closed the active dialog.

Once you have inserted dialog items, the tree shows the structure of your dialog. Changing the order in the structure modifies the layout of the dialog.

To change the order of the tabs in the dialog, select a tab on the tab **Elements**, hold down **Shift** and press the up arrow or the down arrow to move this tab to the left or to the right. The dialog shows the first tab in the structure as the leftmost tab.

Open

To modify an existing dialog item, right-click it and select **Open** on the context menu. This opens the dialog of the **Dialog Item** where you can change it.

Delete

To delete a dialog item, right-click it and select **Delete** on the context menu.

See also

[Insert a Dialog Item into Your Dialog](#)

[Define Properties of a Dialog Item](#)

[Set Parameters for Objects in Your Own Dialog](#)

Inserting a Dialog Item into Your Dialog

Insert a Dialog Item into Your Dialog

Before you start creating your dialog, take a few moments to decide about its structure, i.e., which dialog items do you need, can you show them all in a flat hierarchy in the dialog or do you need to insert tabs and menus.

Remarks

The structure you decide upon determines how and in which order you create the dialog items. Let us say, you want to show the dialog items on tabs. Then, you first have to create the **Tab Control**, which is the container that accommodates the individual tabs. After this you have to create the individual **Tabs**, and finally you can insert the dialog items on each one of these tabs.

- To create a new dialog item, select one of these items on the context menu:

|                      |                           |
|----------------------|---------------------------|
| New Button           | New List View             |
| New Check Box        | New Menu/New Menu Command |
| New DropDownList Box | New Radio Button          |
| New Edit Text Box    | New Static Text Box       |
| New Group Box        | New Tab Control           |
| New Image            | New Tab Page              |
| New List Box         |                           |

- To add one tab at a time to a **Tab Control**, click it with the right mouse button on the tab **Elements**, select **Tab Page** on the context menu, and type in a name and caption into the text box **Dialog Item Tab Page**.
- To add a dialog item to a tab on a **Tab Control** you created, click the name of the **tab** with the right mouse button in the structure on the tab **Elements** and select the item you want to insert on the context menu.

You can type in the same **Name** for dialog items, as long as those dialog items are not located in the same group of items. Groups of items are: **Group Box**, **Menu**, **Tab Control** and **Tab**.

To rearrange the order of items of type **Tab** within a **Tab Control** and **Menu** in the structure on the tab **Elements** within these groups, use **Shift+Up arrow** or **Shift+Down arrow**. The structure shows all other dialog items according to their position in the user-defined dialog itself.

To move items out of one of the groups or place them there, click the item with the left mouse button in the structure, drag it to the new location and drop it there.

SimTalk

[getItemsList \[SimTalk\]](#)

[deleteItem \[SimTalk\]](#)

See also

[Define Properties of a Dialog Item](#)

[Set Parameters for Objects in Your Own Dialog](#)

[Design a Simple Dialog](#)

[Design a Tabbed Dialog](#)

## New Button

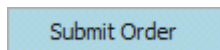
Inserts a **button** into the dialog.

### Image on the Tab Elements



### Description

When the user clicks a button, it calls the **Callback Method** with the parameter you typed in into the text box **Callback Argument**. It in turn executes the actions you programmed.



Proceed as follows:

- Type in the **Name** of the button.
- Type in the **Caption** that the button displays.
- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width**. The default value of 0 automatically sets the width of the button to the width of the **OK** button.
- Select if you want to **Enable** the dialog item or not.
- To insert the dialog item into your dialog, click **OK**.

### SimTalk

**createButton** [SimTalk]

**setCaption** [SimTalk]

**setSensitive** [SimTalk]

See also

**Add a Button**

## New Check Box

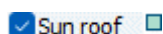
Inserts a **check box** into the dialog. You can also select if the **check box** functions as an **Inheritance box**.

### Image on the Tab Elements



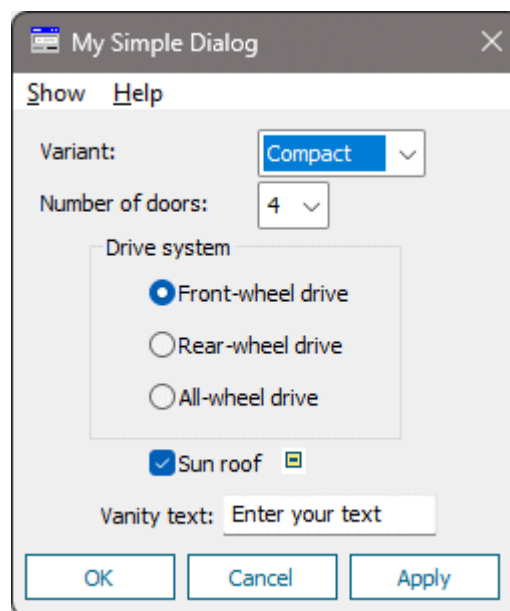
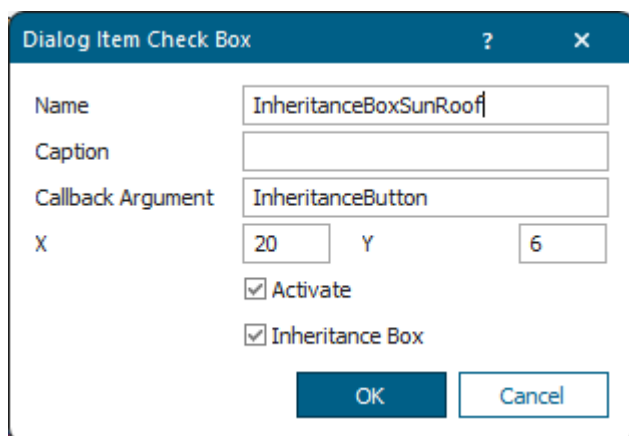
### Description

A check box displays a setting or a set of settings that do not exclude each other. The user can select or clear one or several check boxes at the same time.

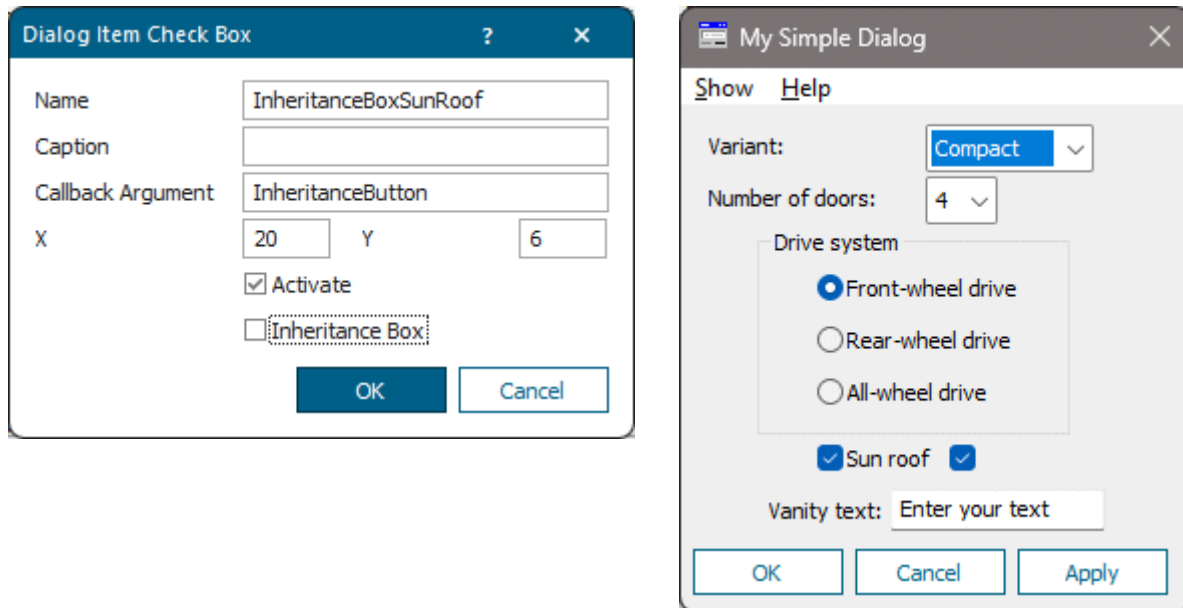


Proceed as follows:

- Type in the **Name** of the check box.
- Type in the **Caption** that describes what the check box does.
- Type in the **Callback Argument**, which will be passed to the **Callback Method** as a parameter.  
It will be executed, when the user selects or clears the **check box**.
- Type in its **X** and **Y** position in the *Dialog*.
- Select if you want to **Enable** the dialog item or not.
- Select the check box **Inheritance box** so that the *Dialog* shows the check box as a check box for inheritance. It looks like a check box for inheritance in the dialogs of the built-in objects. The check box also works like this.



Clear the check box **Inheritance box** so that the *Dialog* shows the check box as a normal check box.



Our **callback method** looks like this:

```
param action: string

switch action
case "NameEdit"
    ?.setCheckBox("NameInherit", false)
    // clear check box when the value changes

case "Open"
    ?.setCaption("NameEdit", ?.NameValue)
    ?.setCheckBox("NameInherit", ?.NameValue.InheritValue)

case "Apply"
    var inherit:boolean := ?.getCheckBox("NameInherit") // 'inherit' is a
    user-defined variable

    if inherit
        ?.NameValue := ?.Origin.NameValue // 'NameValue' is a user-defined
        attribute
        ?.NameValue.InheritValue := true
        ?.setCaption("NameEdit", ?.NameValue)
    else
        ?.NameValue := ?.getValue("NameEdit")
        ?.NameValue.InheritValue := false
    end
end
```

- To insert the dialog item into your dialog, click **OK**.

## SimTalk

[createCheckBox \[SimTalk\]](#)

[getCheckBox \[SimTalk\]](#)

[getInheritanceBox \[SimTalk\]](#)

[getValue \[SimTalk\] - Dialog](#)

[setCaption \[SimTalk\]](#)

[setCheckBox \[SimTalk\]](#)

[setInheritanceBox \[SimTalk\]](#)

[setSensitive \[SimTalk\]](#)

## See also

[Add a Check Box](#)

[Inheritance \[general description\]](#) in the Reference Help

[Using Inheritance](#)



## New DropDownList Box

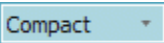
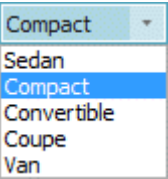
Inserts a **drop-down list box** into the dialog.

Image on the Tab Elements



### Description

The user can select a single item from the drop-down list. When it is closed, it displays the current value. When the user clicks the down arrow , it opens and displays more items for changing the value.

| Collapsed                                                                         | Opened                                                                             |
|-----------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
|  |  |

Proceed as follows:

- Type in the **Name** of the drop-down list.
- Type in the **Callback Argument** which will be passed to the **Callback Method** as a parameter.  
It will be executed when the user closes the **drop-down list box**.
- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width**. The default value of 0 sets the width to the values defined by us.
- Click **Items** and type in the items the list box displays into the dialog **Items**.
- Select if you want to **Enable** the dialog item or not.
- To insert the dialog item into your dialog, click **OK**.

### SimTalk

[createDropDownListBox \[SimTalk\]](#)

[setIndex \[SimTalk\]](#)

[getIndex \[SimTalk\]](#)

[setSensitive \[SimTalk\]](#)

[setList \[SimTalk\]](#)

**See also**

[Add a Drop-down List](#)

New Edit Text Box

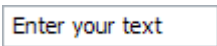
Inserts an **edit text box** into the dialog.

Image on the Tab Elements

**abl**

Description

The user can type text into the edit text box or edit text it already contains.



Proceed as follows:

- Type in the **Name** of the text box.
- Type in the **Callback Argument**, which will be passed to the **Callback Method** as a parameter.

It will be executed when the user changes the contents of the **Text Box** and selects another dialog item afterward, clicks in another text box, or clicks **OK**, **Apply** or **Cancel**.

If the edit text box has a **Callback Argument**, the *Callback Method* of the *Dialog* will be called and the *Callback Method* is assigned the **callback argument \_Enter**, when the user presses the **Enter** key and if the dialog does not show the **Standard Buttons**.

- Select which kind of characters the user can type into the text box.

| Setting               | The user can type in                                                                                                                                                                          |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Any Character         | any character, which includes special characters, lower and upper case letters and numbers.                                                                                                   |
| Letters and Digits    | lower and upper case letters, including special characters, such as umlauts, and letters of any kind of language, underscores, and blank spaces. The user can also type in the digits 0 to 9. |
| Letters               | lower and upper case letters, including special characters, such as umlauts, and letters of any kind of language, and blank spaces.                                                           |
| Decimal Number        | 0123456789                                                                                                                                                                                    |
| Signed Decimal Number | decimal numbers with a negative or positive sign: -+0123456789                                                                                                                                |
| Hexadecimal Number    | ADbf09, for example.                                                                                                                                                                          |
| Octal Number          | 01234567, but not 18 for example.                                                                                                                                                             |
| Binary Number         | 0 or 1                                                                                                                                                                                        |
| Floating Point Number | 12.3E-45, for example.                                                                                                                                                                        |

| Setting              | The user can type in           |
|----------------------|--------------------------------|
| Time                 | time values.                   |
| Date with Time       | a date containing time values. |
| Date                 | a date.                        |
| Positive Real Number | a positive real number.        |

- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width**.
- Select if you want to **Enable** the dialog item or not.
- To show a **password** that the user types into the text box with superscripted lower case Xes , instead of as clear text , select ☒ **Password**.
- To insert the dialog item into your dialog, click **OK**.

SimTalk

[createEditTextBox \[SimTalk\]](#)

[setEditType \[SimTalk\]](#)

[setCaption \[SimTalk\]](#)

[getValue \[SimTalk\] - Dialog](#)

[setSensitive \[SimTalk\]](#)

[setPasswordMasking \[SimTalk\]](#)

[getPasswordMasking \[SimTalk\]](#)

See also

[Add a Text Box](#)

## New Group Box

Inserts a **group box**.

Image on the Tab Elements



### Description

A group box graphically groups a set of dialog items or controls by drawing a frame around it.

| Empty | Populated |
|-------|-----------|
|       |           |

Proceed as follows:

- Type in the **Name** of the group box.
- Type in the **Caption** the group box displays.
- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width** and its **Height**.
- Select if you want to **Enable** the dialog item or not.
- To insert the dialog item into your dialog, click **OK**.

### SimTalk

[createGroupBox \[SimTalk\]](#)

[setCaption \[SimTalk\]](#)

[setSensitive \[SimTalk\]](#)

See also

[Add a Group Box Around Dialog Items](#)

## New Image

Inserts an **image** into the dialog.

### Image on the Tab Elements



### Description

An image is an icon you defined for the object *Dialog*. You can type in a number, or a name, such as **Icon1**.

Proceed as follows:

- Type in the **Name** of the image.
- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width** and **Height** in the *Dialog*.
- Type in the **Image ID**, i.e., the number of the icon, or an image name. The image is an icon of the *Dialog* to which the image belongs.

### SimTalk

**createImage [SimTalk]**

**setIcon [SimTalk]**

**getIcon [SimTalk]**

See also

[Add an Image](#)

## New List Box

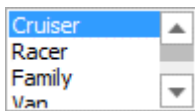
Inserts a **list box** into the dialog.

### Image on the Tab Elements



### Description

A list box displays a list of choices. The user can double-click one of the items in the list to select it. The list does not collapse.



Proceed as follows:

- Type in the **Name** of the list box.
- Type in the **Callback Argument** which will be passed to the **Callback Method** as a parameter.  
It will be executed, when the user selects and double-clicks an item in the list box.
- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width** and **Height** in the *Dialog*. If you do not type in values, the *Dialog* sets the **width** and **height** to the values defined by us. When you type in a value for the **height**, it adds a vertical scrollbar to the list if the list contains more items than the **height**.
- Click **Items** and type in the items the list box displays into the dialog **Items**.
- Select if you want to **Enable** the dialog item or not.
- To insert the dialog item into your dialog, click **OK**.

## SimTalk

### Mutation [SimTalk] - GASequence

### memUsage [SimTalk]

### getIndex [SimTalk]

### setList [SimTalk]

### setSensitive [SimTalk]

### getValue [SimTalk] - Dialog

See also

[Add a List Box](#)[Add a List Box](#)



## New List View

Inserts a **list view** into the dialog.

### Icon on the Tab Elements



### Description

A list view shows a table in the dialog.

| Model | Variant | Color  | Accessories    |
|-------|---------|--------|----------------|
| Racer | Coupe   | silver | mag wheels     |
|       |         |        | curb feelers   |
|       |         |        | surround sound |
|       |         |        |                |
|       |         |        |                |

#### Note

When displaying *strings*, *Plant Simulation* only shows the first 260 characters.

Proceed as follows:

- Type in the **Name** of the list view.
- Type in the **Callback Argument** which will be passed to the **Callback Method** as a parameter. It will be executed, when the user double-clicks a row in the **list view**.

#### Note

The user can only select rows, but no columns or individual cells.

- Type in the name of a table or click  and select a table in the dialog **Select Object**.  
Type the items the list view shows into the cells of the columns.
- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width** and **Height** in the *Dialog*.
- Select if you want to **Enable** the dialog item or not.
- To insert the dialog item into your dialog, click **OK**.

## SimTalk

[createListView \[SimTalk\]](#)

[setTableRow \[SimTalk\]](#)

[getTableRow \[SimTalk\]](#)

[setSensitive \[SimTalk\]](#)

[getTable \[SimTalk\] - Dialog](#)

[setTable \[SimTalk\] - Dialog](#)

**See also**

[Add a List View](#)

## New Menu/New Menu Command

Inserts a **menu** or a **menu command** into the dialog.

### Image on the Tab Elements



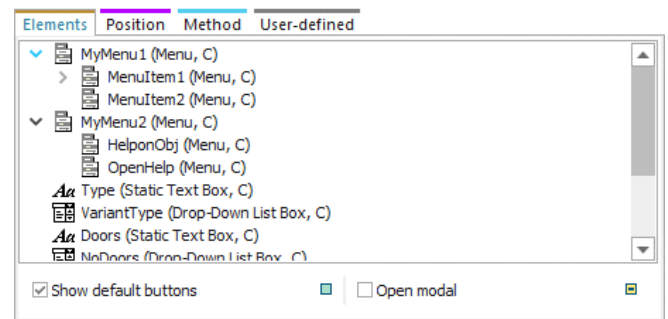
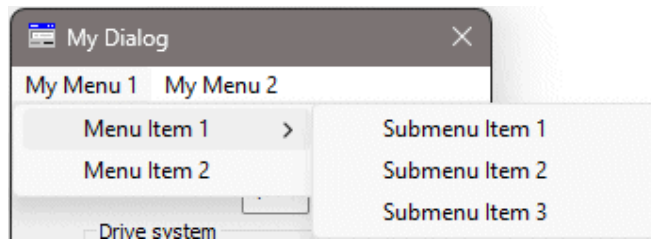
### Description

A menu provides a list of textual choices from which the user can choose. The *Dialog* inserts the menus along the top of the dialog you define.

Proceed as follows:

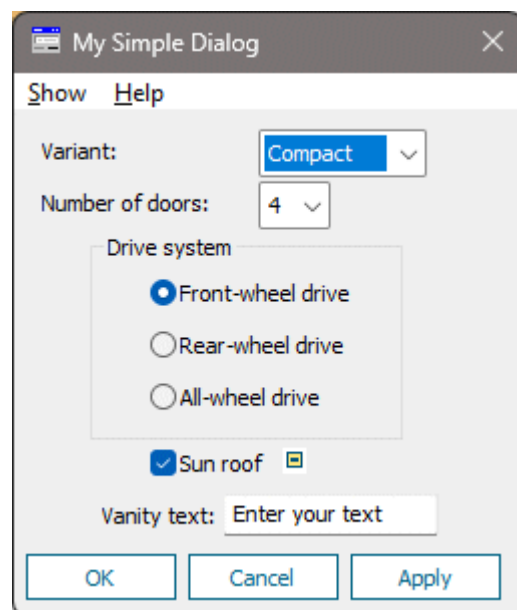
- Type in the **Name** of the menu.
- Type in the **Caption** of the menu. Type in a hyphen (-) as the caption of the menu command to insert a menu separator between groups of menu commands.
- Type in the **Callback Argument** which will be passed to the **Callback Method** as a parameter.  
It will be executed when the user selects a **menu** or a **menu command**.
- Select if you want to **Enable** the dialog item or not.
- To insert the menu into your dialog, click **OK**.
- To add one menu command at a time to the selected menu, right-click it and select **New Menu/New Menu Command** again.
  - Type in the **Name** of the menu command.
  - Type in the **Caption** of the menu command.
  - Type in the **Callback Argument** which will be passed to the **Callback Method** as a parameter.
  - Select if you want to **Enable** the dialog item or not.
  - To insert the menu into your dialog, click **OK**.
- To add a **submenu** to the selected menu command, right click that menu command and select **New Menu/New Menu Command** again. Repeat the steps described above.

A submenu looks like this.

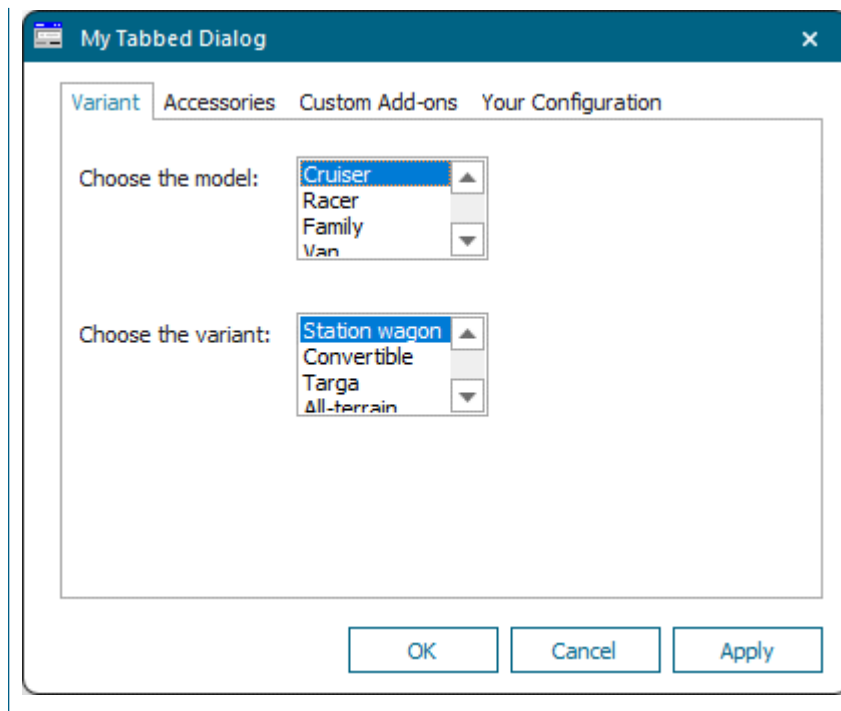


### Note

The **display window** of the object *Dialog* that contains menus does not use the standard Siemens PLM theme, but your Windows theme.



The **display window** of the object *Dialog* that does not contain menus uses the standard Siemens PLM theme.



## SimTalk

[createMenu \[SimTalk\]](#)

[setCaption \[SimTalk\]](#)

[setSensitive \[SimTalk\]](#)

See also

[Add a Menu and Menu Commands](#)

## New Radio Button

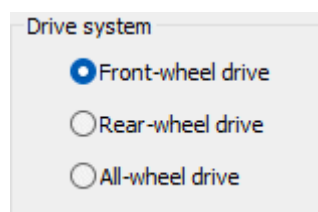
Inserts a **radio button** into the dialog.

### Image on the Tab Elements



### Description

A set of radio buttons enables the user to select a single setting from a fixed set of choices that exclude each other.



Proceed as follows:

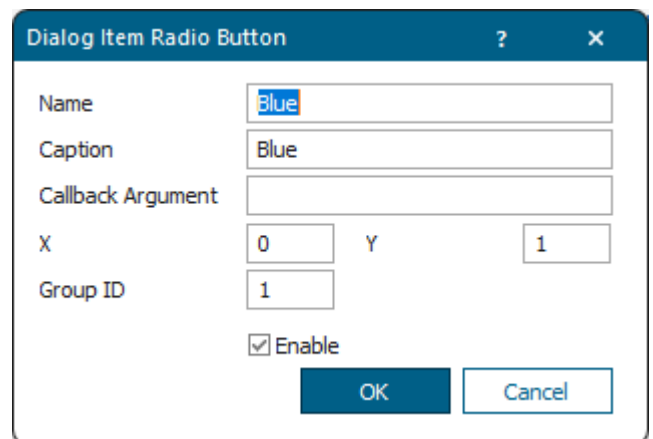
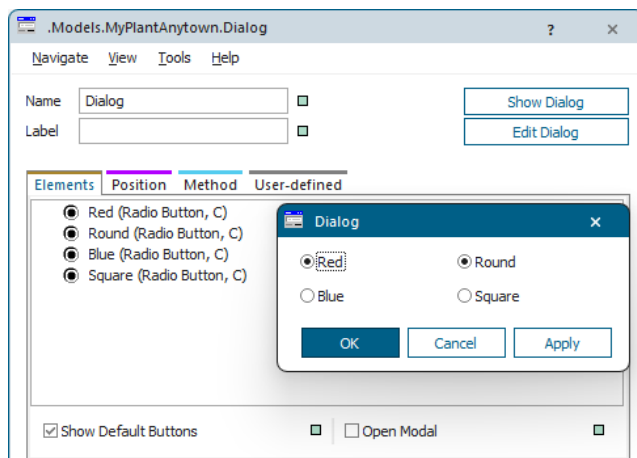
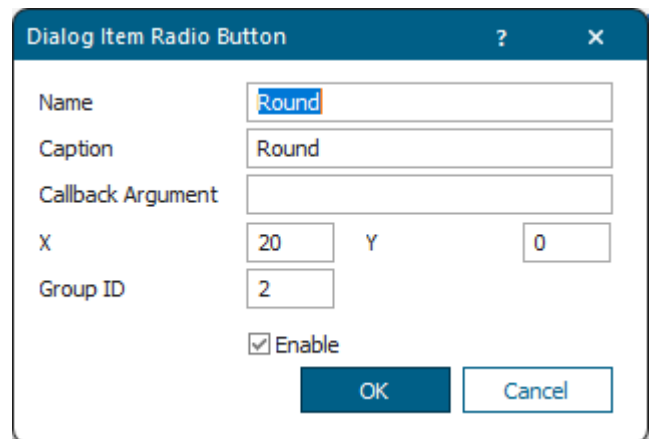
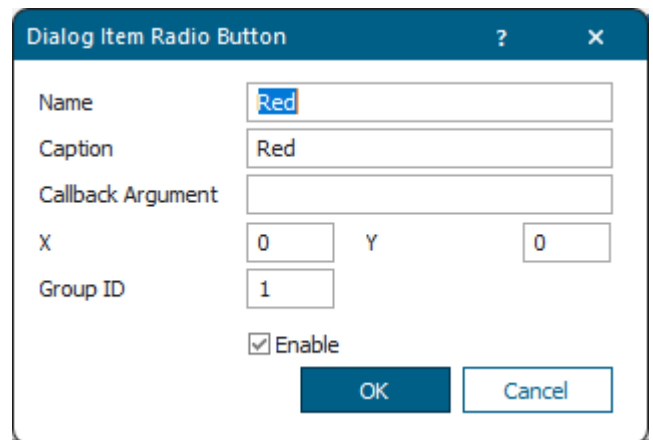
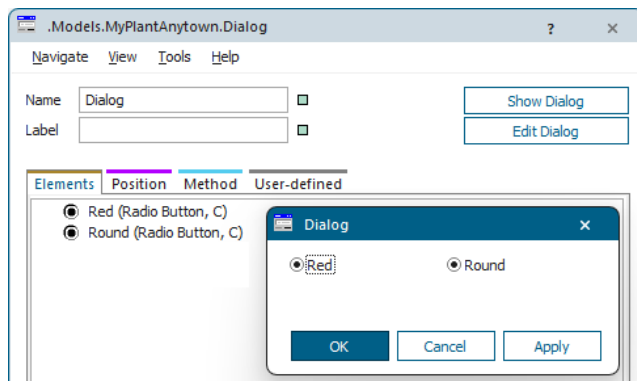
- Type in the **Name** of the radio button.
- Type in the **Caption** that describes what the radio button does and that the *Dialog* shows.
- Type in the **Callback Argument** which will be passed to the **Callback Method** as a parameter.  
It will be executed, when the user selects the **radio button**.
- Type in its **X** and **Y** position in the *Dialog*.
- Type in a **Group ID**, i.e., a number, for grouping any number of radio buttons to denote which radio buttons belong together.

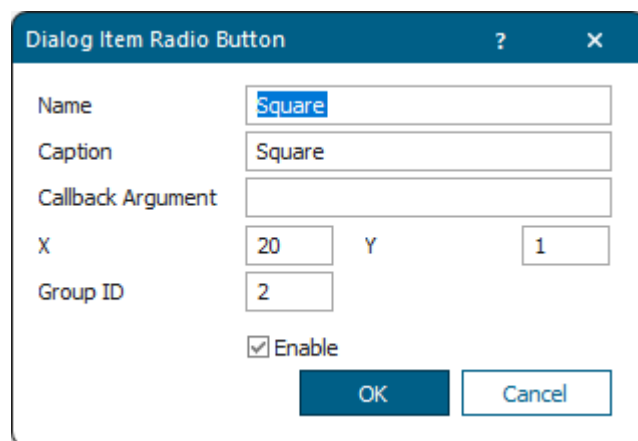
**Group ID 1** might stand for one set of radio buttons, while **Group ID 2** might stand for another set. The user can only select one radio button of a group.

- Select if you want to **Enable** the dialog item or not.

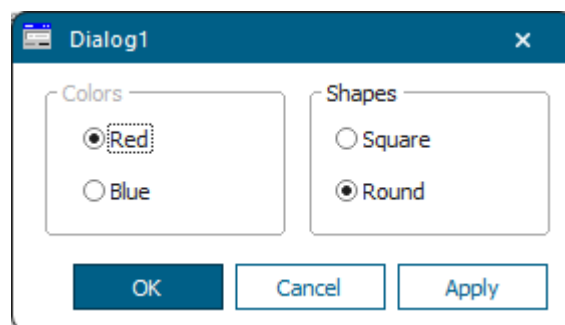
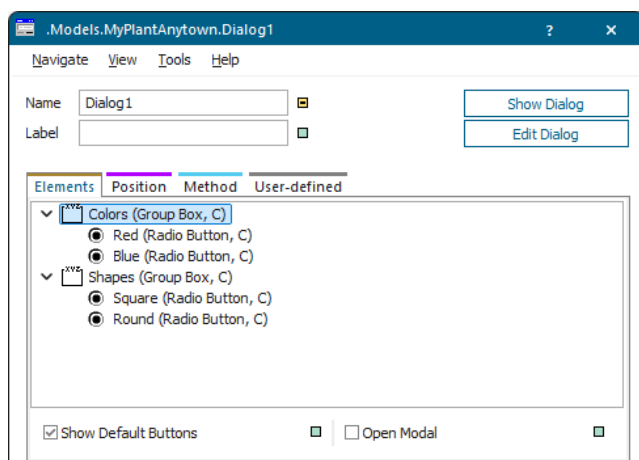
#### Note

You have to create all items of a group of radio buttons sequentially because *Plant Simulation* creates the items according to their row and column position. You cannot arrange two sets of radio buttons as shown below, where the radio buttons **Red** and **Blue** on the left-hand side and the radio buttons **Round** and **Square** on the right-hand side form a group each. Windows ignores the **Group ID** you type in and groups the radio buttons row by row according to the **Y-coordinate**, i.e., it treats **Red** and **Round** as one group and **Blue** and **Square** as the other group.





If you want to model like this, do not place the radio buttons loosely onto the dialog, but group them within group boxes.



- To insert the dialog item into your dialog, click **OK**.

## SimTalk

[createRadioButton \[SimTalk\]](#)

[setCheckBox \[SimTalk\]](#)

[getCheckBox \[SimTalk\]](#)

[setCaption \[SimTalk\]](#)

[setGroupID \[SimTalk\]](#)

[setSensitive \[SimTalk\]](#)



[getValue \[SimTalk\] - Dialog](#)

**See also**

[Add a Set of Radio Buttons](#)

## New Static Text Box

Inserts a **static text box** into the dialog.

### Image on the Tab Elements



### Description

A static text box shows text that the user can view, but not edit.

Vanity text:

Proceed as follows:

- Type in the **Name** of the static text box.
- Type in the **Caption** that the static text box shows.
- Type in its **X** and **Y** position in the *Dialog*.
- Select if you want to **Enable** the dialog item or not.
- To insert the dialog item into your dialog, click **OK**.

### SimTalk

[createStaticTextBox \[SimTalk\]](#)

[setCaption \[SimTalk\]](#)

[setSensitive \[SimTalk\]](#)

See also

[Add a Static Text Box](#)

## New Tab Control

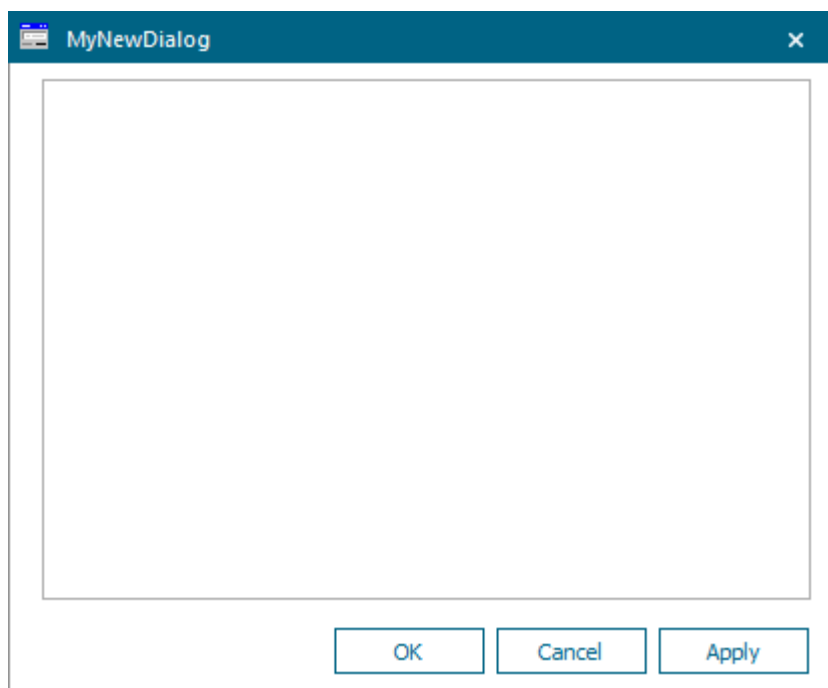
Inserts a **tab control** into the dialog.

### Image on the Tab Elements



### Description

A **tab control** resembles a notebook that contains any number of pages.



The user can navigate between different tabs and/or sections of information within the dialog box.

#### Note

Do not confuse the tab control with a single tab. The tab control is the container that holds the individual tabs (pages).

When the user selects another tab, the *Dialog* calls the **Callback Method**, which in turn executes the actions you programmed with the first parameter you typed in as **Callback Argument**.

Proceed as follows:

- Type in the **Name** of the tab control.
- Type in the **Callback Argument** which will be passed to the **Callback Method** as a parameter.

It will be executed when the user selects a tab in the **tab control**.

- Type in its **X** and **Y** position in the *Dialog*.
- Type in its **Width** and **Height** in the *Dialog*.
- To insert the **tab control** into your dialog, click **OK**.
- To add one **tab** at the time to the **tab control**, click it with the right mouse button on the tab **Elements** and select **New Tab Page** on the context menu.

**Note**

To change the order of the tabs in the dialog, select a tab on the tab **Elements**, hold down **Shift** and press the **up arrow** or the **down arrow** to move this tab to the left or to the right. The dialog shows the first tab in the structure as the leftmost tab.

See also

[createTabControl \[SimTalk\]](#)

[setTab \[SimTalk\]](#)

[memUsage \[SimTalk\]](#)

[getIndex \[SimTalk\]](#)

[getValue \[SimTalk\] - Dialog](#)

[setList \[SimTalk\]](#), applies for all tab pages of the tab control.

See also

[Add a Tab Control](#)

## New Tab Page

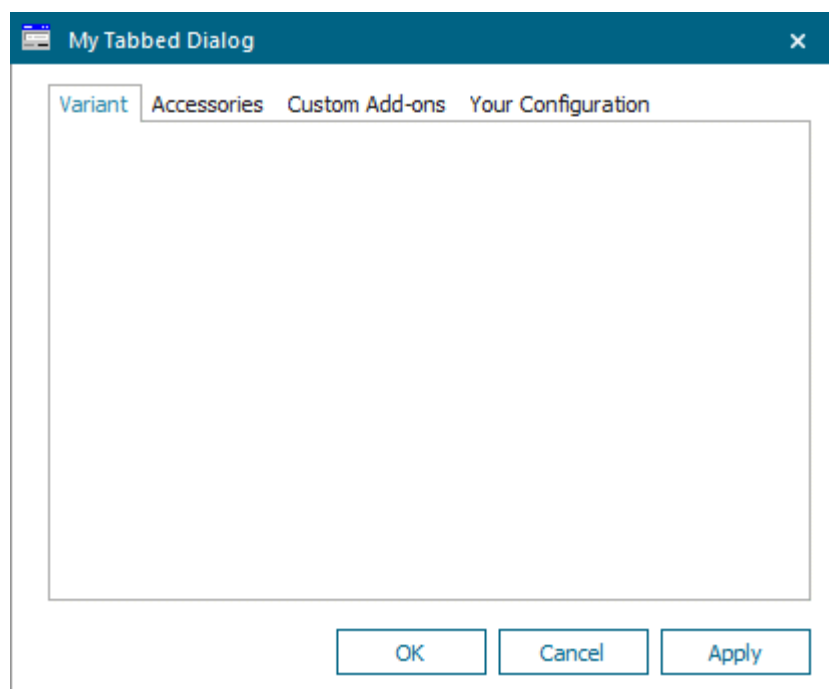
To insert a **tab** at the time to the **tab control**, click it with the right mouse button on the tab **Elements** and select **New Tab Page**.

### Image on the Tab Elements



### Description

A **tab** offers settings that belong together.



Proceed as follows:

- Type in the **Name** of the tab into the dialog named **Dialog Item Tab Page**.
- Type in the **Caption** of the tab into the dialog named **Dialog Item Tab Page**. The tab on the tab control shows this caption as the title on the tab itself.

#### Note

To change the order of the tabs in the dialog, select a tab on the tab **Elements**, hold down **Shift** and press the **up arrow** or the **down arrow** to move this tab to the left or to the right. The dialog shows the first tab in the structure as the leftmost tab.

SimTalk

[createTabPage \[SimTalk\]](#)

[setCaption \[SimTalk\]](#) applies, when you change the caption of one tab page only.

[setTab \[SimTalk\]](#)

See also

[Add Tabs to a Tab Control](#)

Defining Properties of a Dialog Item

Define Properties of a Dialog Item

Depending on the type of item you define, the *Dialog* shows one or all of these items in the dialog of the item you are defining:

|                   |          |
|-------------------|----------|
| Name              | Image ID |
| Caption           | Width    |
| Callback Argument | Height   |
| X                 | Enable   |
| Y                 | Items    |
| Group ID          |          |

## Name [dialog item in Dialog]

Type in the **Name** of the dialog item. The *Dialog* shows this **Name** in the text box on the tab **Elements**.

Name

SunRoof

### Remarks

You can type in the same **Name** for dialog items, as long as those dialog items are not located in the same group of items. Groups of items are: **Group Box**, **Menu**, **Tab Control** and **Tab**.

We recommend to type in a unique identifier, containing the type of dialog item, for example `xyz_button`, `xyz_list`, etc.

#### Note

You cannot use a period (.) in the name of a dialog item. As *SimTalk* treats this name as a string, you can also type in special characters.

A *Method* object can call this dialog item using the **Name** you typed in. For some dialog items you can type in a **Caption** in addition.

### SimTalk

[setCaption \[SimTalk\]](#)

[getValue \[SimTalk\] - Dialog](#)

## Caption [text box] - Dialog

Certain dialog items show the text box **Caption**. Type in the text which the *Dialog* shows as the **caption** of the dialog item in your dialog.

### Remarks

Caption

Sun roof

A **caption** is especially handy if you develop libraries in several languages and want the user to be able to switch languages. In addition, you can type in special characters and blanks, which you cannot use in the name.

The **Caption** applies to these dialog items:

[New Static Text Box](#)

[New Button](#)

[New Group Box](#)

[New Check Box](#)

[New Menu/New Menu Command](#)

[New Tab Page](#)

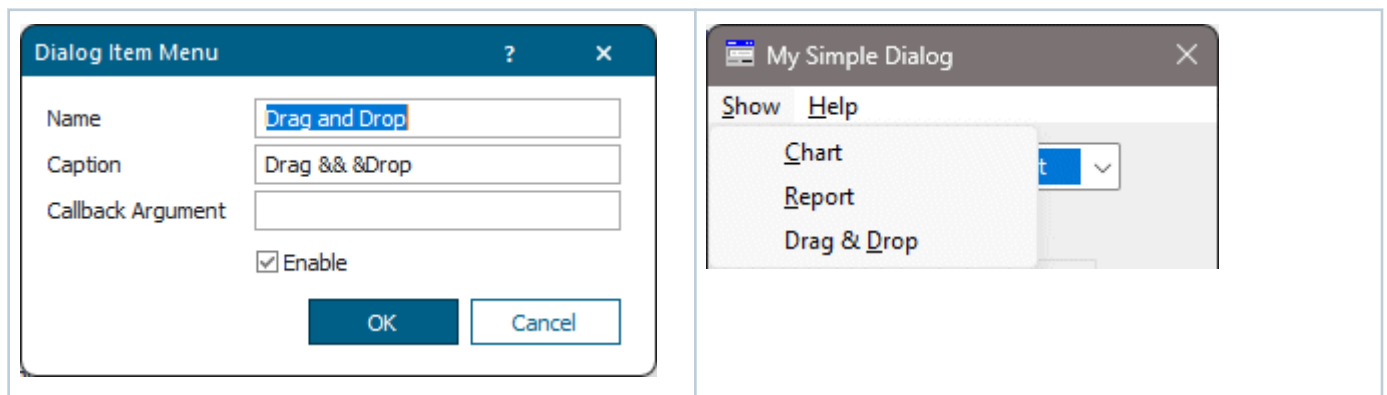
[New Radio Button](#)

### Create an Access Key

Type in the ampersand character **&** in front of a letter of the **Caption** of the **dialog items** listed above to make this letter the access key (mnemonic key). *Plant Simulation* activates this dialog item, when you press **Alt** and that letter.

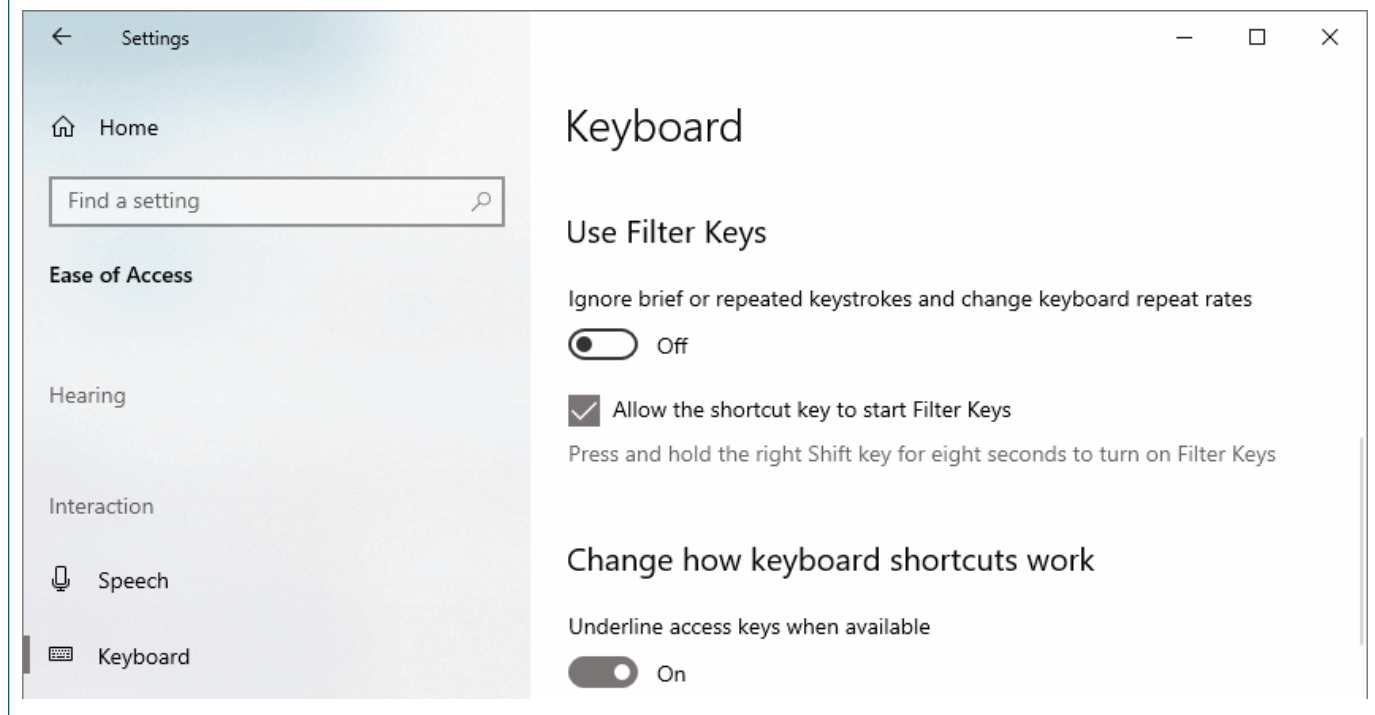
If you want to use the ampersand character **&** instead of **and** in a dialog item, for example **Drag & Drop**, type in the **&** twice, Drag **&&** Drop.





### Note

If your dialogs do not show the access keys, make sure that **Settings > Ease of Access > Keyboard > Change how keyboard shortcuts work** is **On**.



### SimTalk

[setCaption \[SimTalk\]](#)

[getValue \[SimTalk\] - Dialog](#)

## Callback Argument [text box] - Dialog

A number of dialog elements show the text box **Callback Argument**.

Callback Argument

### Remarks

Type in the parameter that is passed to the **Callback Method**. Be aware that the parameters are case-sensitive!

The *callback method* executes this parameter when the user performs the respective action, when he:

- Closes the **DropDownList Box**.
- Selects and double-clicks an item in the **List Box**.
- Changes the contents of the **Text Box** and selects another dialog item afterward, clicks in another text box, or clicks **OK**, **Apply**, or **Cancel**.

If the edit **text box** has a **Callback Argument**, the *Callback Method* of the *Dialog* will be called and the *Callback Method* is assigned the **callback argument\_Enter**.

- Clicks the **Button**.
- Selects or clears the **Check Box**.
- Selects the **Radio Button**.
- Selects a row in the **List View** and double-clicks it.
- Selects a tab in the **Tab Control**.
- Selects a **Menu/Menu Command**.

### SimTalk

#### setCallbackArgument [SimTalk]

### See also

#### Program Actions which the Dialog Items Execute

## X [text box] - Dialog

Type in the **position on the x-axis** where *Plant Simulation* places the dialog item in the *Dialog* you define. The value corresponds to the average width of characters of your system font, not to pixels.

X  Y

### Remarks

If you cannot see the dialog items you inserted, check the settings for the position first. If you typed in identical values for different dialog items, the *Dialog* places them on top of each other, so that you can only see the first item(s) you inserted.

### SimTalk

[DialogX \[SimTalk\]](#)

[DialogY \[SimTalk\]](#)

## Y [text box] - Dialog

Type in the **position on the y-axis** where *Plant Simulation* places the dialog item in the *Dialog* you define. The value corresponds to lines of characters of the system font, not to pixels. You can type in a value from 0 to 10 for example.

X  Y

## Remarks

If you cannot see the dialog items you inserted, check the settings for the position first. If you typed in identical values for different dialog items, the *Dialog* places them on top of each other, so that you can only see the first item(s) you inserted.

## SimTalk

[DialogY \[SimTalk\]](#)

[DialogX \[SimTalk\]](#)

## Group ID

For a dialog item of **Type > Radio Button** the *Dialog* shows the text box **Group ID**. Type in a **Group ID**, i.e., a number, for grouping any number of radio buttons to denote which radio buttons belong together.

Group ID

### Remarks

**Group ID** 1 might stand for one set of radio buttons, while **Group ID** 2 might stand for another set. The user can only select one radio button of a group.

### SimTalk

[setGroupID \[SimTalk\]](#)

### See also

[New Radio Button](#)

## Image ID

For a dialog item of **Type > Image** the *Dialog* shows the text box **Image ID**. Type in an **Image ID**, i.e., the number of the icon, or the name of the image. The image is an icon of the *Dialog* to which the image belongs.

Image ID

Icon3

## SimTalk

[createImage \[SimTalk\]](#)

[setIcon \[SimTalk\]](#)

[getIcon \[SimTalk\]](#)

See also

[New Image](#)

## Width [text box] - dialog item

Type the **Width** of the dialog item in the average width of characters of your system font into the text box.  
The default value of **0** sets the width to the values defined by us.

Width

Height

## Height [text box] - dialog item

Type in the **Height** of the dialog item in lines of characters of your system font. The default value of **0** sets the height to the values defined by us.

Width

Height



## Enable [check box] - dialog item

Select this check box to **Enable** the respective dialog item. Clear the check box to not enable it.

☒ Enable

### Remarks

For a dialog item of **Type** > **New Button**, **New Check Box**, **New Radio Button**, **New Edit Text Box**, **New List Box**, and **New List View** the *Dialog* shows the check box **Enable**.

You can:

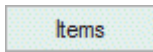
- Select **Enable** to activate the respective dialog item, **Vanity text:** for example, in the *Dialog*. When the user selects it, it triggers the **Callback Method**, which executes the actions you programmed.
- Clear it to deactivate the dialog item, i.e., to gray it out **Vanity text:** in the *Dialog*, and thus make it not available to the user.

### SimTalk

**setSensitive [SimTalk]**

## Items [Dialog]

The *Dialog* shows the button **Items** for a dialog item of **Type > DropDown List Box** and **ListBox**.



### Remarks

Click **Items** to open a dialog into which you type the items they show.

- To add an item to the list field, type its name into the text box and click **Insert** or press the **Enter** key.  
Repeat this for all the items you want to add.
- To delete an item from the list, select it in the list field and click **Delete**.
- To move an item up one position in the list, select it in the list field and click **Move Up**.
- To change the name of an item in the list, select it in the list field, type another name into the text box, and click **Rename**.

To apply the items you typed in and to close the dialog, click **OK**.

To discard what you have done so far, click **Cancel**.

### SimTalk

[setIndex \[SimTalk\]](#)

[getIndex \[SimTalk\]](#)

### See also

[DropDownList Box](#)

[List Box](#)

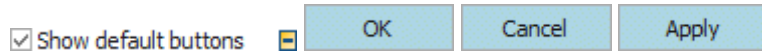
[setList \[SimTalk\]](#)

[setCaption \[SimTalk\]](#)

[getValue \[SimTalk\] - Dialog](#)

## Show Default Buttons [check box]

To show the **standard buttons** in the dialog that you are creating, select this check box. To hide these buttons, clear the check box.



### Remarks

If you do not show the button combination **OK**, **Cancel**, and **Apply**, you have to define your own buttons to provide the user with a way to apply or to discard settings he chose.

### SimTalk

#### ShowStandardButtons [SimTalk]

## Open Modal [check box]

To open the dialog box you are creating **modal**, select this check box. Then, the user cannot open any other *Plant Simulation* dialog boxes until he closes the dialog box of the user-defined dialog. To allow the user to open other dialog boxes in addition to the dialog box you are creating, clear the check box.

☒ Open modal 

SimTalk

[OpenModal \[SimTalk\]](#)

## Tab Position

### Tab Position

The **Tab Position** provides the settings listed in the table of contents to the left.

## X-Position [dialog item]

Type in the position on the x-axis of your computer screen at which *Plant Simulation* shows the *Dialog* when you open it.

X-Position   Y-Position  

### Remarks

The default setting **-1** for both **X-Position** and **Y-Position** centers the *Dialog* on screen.

The unit is pixels, the zero point is the top left corner of the screen.

### SimTalk

[getUserDialogXYWH \[SimTalk\]](#)

## Y-Position [dialog item]

Type in the position on the y-axis of your computer screen at which *Plant Simulation* shows the *Dialog* when you open it.

X-Position  Y-Position

### Remarks

The default setting **-1** for both **X-Position** and **Y-Position** centers the *Dialog* on screen.

The unit is pixels, the zero point is the top left corner of the screen.

### SimTalk

[getUserDialogXYWH \[SimTalk\]](#)

### Tab Method

### Tab Method

The **Tab Method** provides the settings listed in the table of contents to the left.

The **parameters** are going to be called when *Plant Simulation* executes the **Callback Method**.

## Callback Method [text box] - Dialog

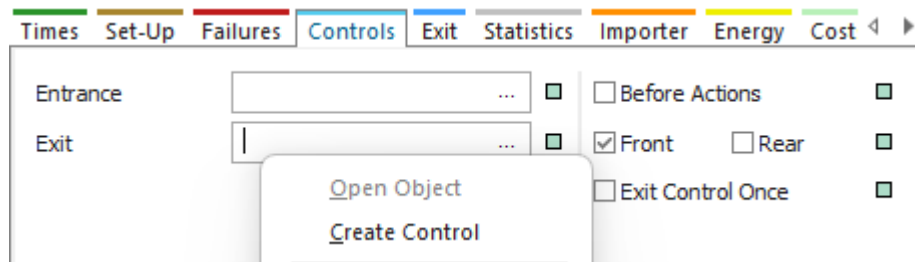
Creates a **Callback Method** for the *Dialog*.

|                    |                                                         |
|--------------------|---------------------------------------------------------|
| Callback Method    | <code>self.Callback</code> ... <input type="checkbox"/> |
| Argument for Open  | Open <input type="checkbox"/>                           |
| Argument for Apply | Apply <input type="checkbox"/>                          |
| Argument for Close | Close <input type="checkbox"/>                          |

### Remarks

Proceed as follows to create the **Callback Method**:

- Click  and select a *Method* in the dialog box **Select Object**. Type in the source code of the callback method. It sets the actions which the dialog items execute. Or you can accept the default, namely the *user-defined attribute* of data type *method* named `self.callback`.
- Select the *Method* in a *Frame*, drag it to the text box and drop it there.
- Select **Create Control** and create the control as a *Method of the object*.



Type in the source code of this control into the *Method* that opens.

To edit the source code later on:

- Press **F2**.
- Or hold down **Shift** and double-click into the text box.
- Or click the tab **User-defined** and double-click the name of the *Method* in the list.

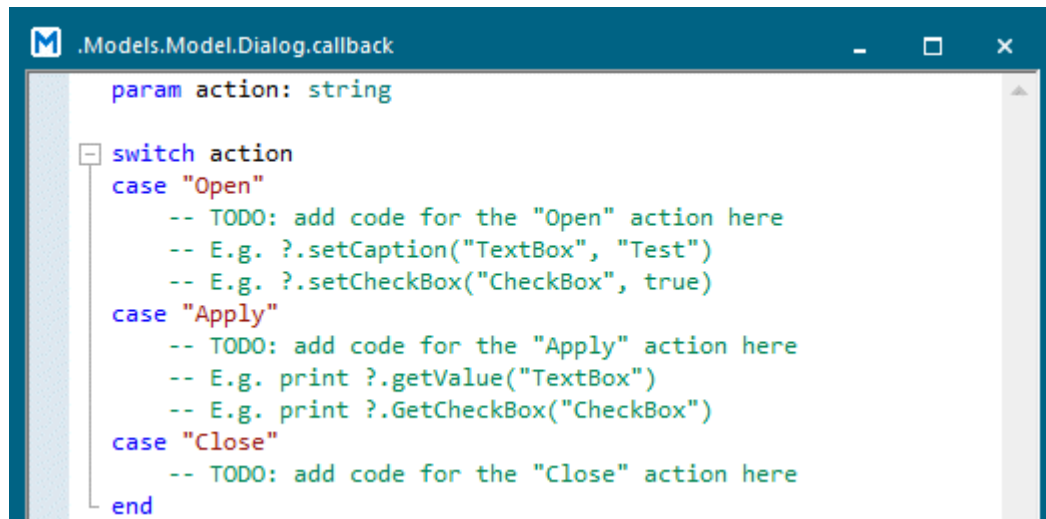
### Parameters

The *callback method* can have the following parameters:

- The **Open** section initializes the contents of the dialog box or sets the dialog items to values of your choice.
- The *Dialog* executes the **Apply** section when the user clicks **OK** or **Apply** in the dialog box of the *Dialog* you define. The source code you type in may evaluate new or changed values.

- The *Dialog* executes the **Close** section when the user clicks **Cancel** in the dialog box of the *Dialog* you define, or when he closes it with **Close**  on the title bar.
- The *Dialog* executes the **Callback Argument** of a **drop-down List Box** when the user closes it.
- The *Dialog* executes the **Callback Argument** of a **List Box** when the user selects and double-clicks an item in it.
- The *Dialog* executes the **Callback Argument** of a **Text Box** when the user changes its contents and selects another dialog item afterward, clicks in another text box, or clicks **OK**, **Apply**, or **Cancel**.
- The *Dialog* executes the **Callback Argument** of a **Button** when the user clicks the **Button**.
- The *Dialog* executes the **Callback Argument** of a **Check Box** when the user selects or clears it.
- The *Dialog* executes the **Callback Argument** of a **Radio Button** when the user selects it.
- The *Dialog* executes the **Callback Argument** of a **List View** when the user selects a row in it and double-clicks it.
- The *Dialog* executes the **Callback Argument** of a **Tab Control** when the user selects a tab.
- The *Dialog* executes the **Callback Argument** of a **Menu/Menu Command** when the user selects a menu or a menu command.

The **standard callback method** as a *user-defined attribute* looks like this:



```

param action: string

switch action
case "Open"
  -- TODO: add code for the "Open" action here
  -- E.g. ?.setCaption("TextBox", "Test")
  -- E.g. ?.setCheckBox("CheckBox", true)
case "Apply"
  -- TODO: add code for the "Apply" action here
  -- E.g. print ?.getValue("TextBox")
  -- E.g. print ?.GetCheckBox("CheckBox")
case "Close"
  -- TODO: add code for the "Close" action here
end
  
```

SimTalk

[CallbackMethod \[SimTalk\] - Dialog](#)

See also

[Program Actions which the Dialog Items Execute](#)

[Callback Argument \[text box\] - Dialog](#)



## Argument for Open [text box]

Type in the name of the callback argument for the **Open** action into the text box.

|                    |                                            |                          |
|--------------------|--------------------------------------------|--------------------------|
| Callback Method    | <input type="text" value="self.Callback"/> | <input type="checkbox"/> |
| Argument for Open  | <input type="text" value="Open"/>          | <input type="checkbox"/> |
| Argument for Apply | <input type="text" value="Apply"/>         | <input type="checkbox"/> |
| Argument for Close | <input type="text" value="Close"/>         | <input type="checkbox"/> |

### Remarks

The *Dialog* executes the **Argument for Open** when it executes the **Callback Method**. The default setting is **Open**.

The **Open** section of the *callback method* initializes the contents of the dialog box or sets the dialog items to values of your choice.

### SimTalk

#### ArgumentForOpen [SimTalk]

### See also

#### Program Actions for Interacting with the Dialog

## Argument for Apply [text box]

Type in the name of the callback argument for the **Apply** action into the text box.

|                    |                                            |                          |
|--------------------|--------------------------------------------|--------------------------|
| Callback Method    | <input type="text" value="self.Callback"/> | <input type="checkbox"/> |
| Argument for Open  | <input type="text" value="Open"/>          | <input type="checkbox"/> |
| Argument for Apply | <input type="text" value="Apply"/>         | <input type="checkbox"/> |
| Argument for Close | <input type="text" value="Close"/>         | <input type="checkbox"/> |

### Remarks

The *Dialog* executes the **Argument for Apply** when it executes the **Callback Method**. The default setting is **Apply**.

The *Dialog* executes the **Apply** section of the *callback method* when the user clicks **OK** or **Apply** in the dialog box of the *Dialog* you define. The source code that you type in may evaluate new or changed values.

#### Note

When the user clicks **OK**, the *Dialog* executes the *callback method* twice. The first time the **Apply** section is called. The second time the **Close** section is called.

When the user clicks **Apply**, the *Dialog* only executes the **Apply** section of the *callback method*.

### SimTalk

#### ArgumentForApply [SimTalk]

#### See also

#### Program Actions for Interacting with the Dialog

## Argument for Close [text box]

Type in the name of the callback argument for the **Close** action into the text box.

|                    |                                            |                          |
|--------------------|--------------------------------------------|--------------------------|
| Callback Method    | <input type="text" value="self.Callback"/> | <input type="checkbox"/> |
| Argument for Open  | <input type="text" value="Open"/>          | <input type="checkbox"/> |
| Argument for Apply | <input type="text" value="Apply"/>         | <input type="checkbox"/> |
| Argument for Close | <input type="text" value="Close"/>         | <input type="checkbox"/> |

### Remarks

The *Dialog* executes the **Argument for Close** when it executes the **Callback Method**. The default setting is **Close**.

The *Dialog* executes the **Close** section of the *callback method* when the user clicks **Cancel** in the dialog box of the *Dialog* you define, or when he closes it with the **Close** button  on the title bar.

#### Note

When the user clicks **OK**, the *Dialog* executes the *callback method* twice. The first time the **Apply** section is called. The second time, the **Close** section is called.

When the user clicks **Apply**, the *Dialog* only executes the **Apply** section of the *callback method*.

### SimTalk

#### ArgumentForClose [SimTalk]

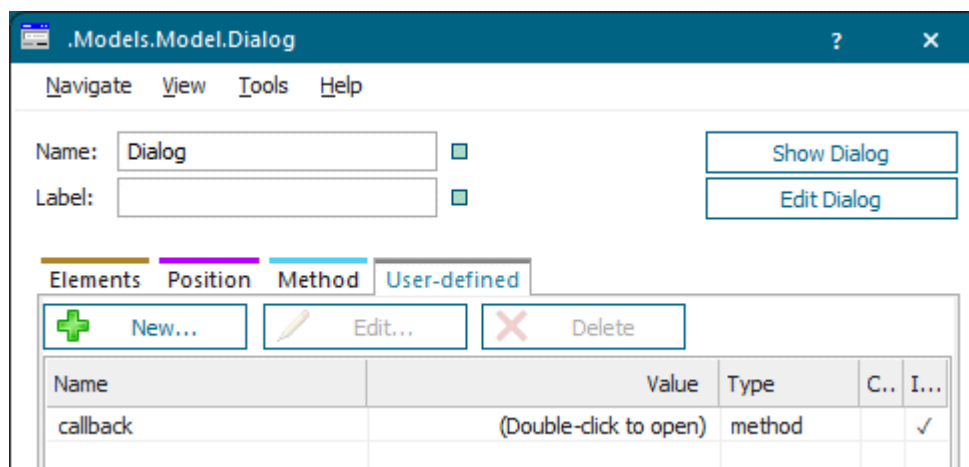
#### See also

#### Program Actions for Interacting with the Dialog

## Tab User-defined

Define your own attributes as described under the [Tab User-defined](#).

The *Dialog* provides the [Callback method](#) as a *user-defined attribute*.



## Navigate Menu

The commands are described under the [Navigate Menu](#).

## View Menu

The commands are described under the [View Menu](#).

### SimTalk

[updateDialog \[SimTalk\]](#)

## Tools Menu

The **Tools Menu** provides commands to access its functions.

[Edit Controls](#)

[Edit Observers](#)

See also

[Tools Menu \[general description\]](#)

## Help Menu

The commands are described under the [Help Menu](#).

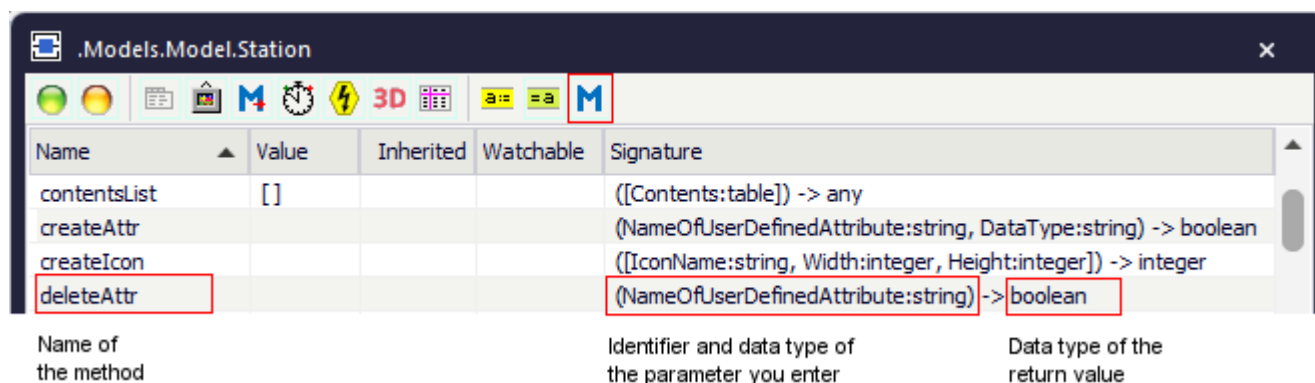
## Methods of the Dialog

### \_Methods of the Dialog

The *Dialog* provides:

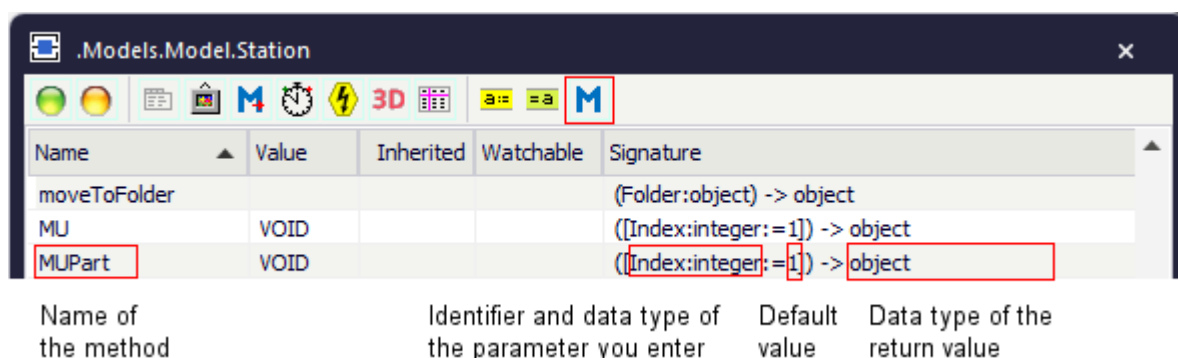
- The *methods* listed in the table of contents to the left.
- The [Methods of All Objects](#).

To view all of the *methods*, *read-only attributes*, and *attributes* of the object, open the window [Show Attributes and Methods](#). The figure below illustrates the information using the example of the object *Station*.



| Name         | Value | Inherited | Watchable | Signature                                                       |
|--------------|-------|-----------|-----------|-----------------------------------------------------------------|
| contentsList | []    |           |           | [[Contents:table]] -> any                                       |
| createAttr   |       |           |           | (NameOfUserDefinedAttribute:string, DataType:string) -> boolean |
| createIcon   |       |           |           | ([IconName:string, Width:integer, Height:integer]) -> integer   |
| deleteAttr   |       |           |           | (NameOfUserDefinedAttribute:string) -> boolean                  |

Name of the method                      Identifier and data type of the parameter you enter                      Data type of the return value



| Name         | Value | Inherited | Watchable | Signature                      |
|--------------|-------|-----------|-----------|--------------------------------|
| moveToFolder |       |           |           | (Folder:object) -> object      |
| MU           | VOID  |           |           | ((Index:integer:=1)) -> object |
| MUPart       | VOID  |           |           | ((Index:integer:=1)) -> object |

Name of the method                      Identifier and data type of the parameter you enter                      Default value                      Data type of the return value

- Select **Show Attributes and Methods** on the context menu of the **Class Library** to show the *methods*, *read-only attributes*, and *attributes* of the selected [Class \[general description\]](#).
- Press the **F8** key or click **Show Attributes and Methods** on the **Home** ribbon tab of the *Frame* into which you inserted an instance to show the *methods*, *read-only attributes*, and *attributes* of the selected [Instance \[general description\]](#).